

Figure 1 shows the loads acting on a driven shaft made of a grey cast iron (brittle material, $R_m = 600 \text{ MPa}$) due to a helicoidal gear.

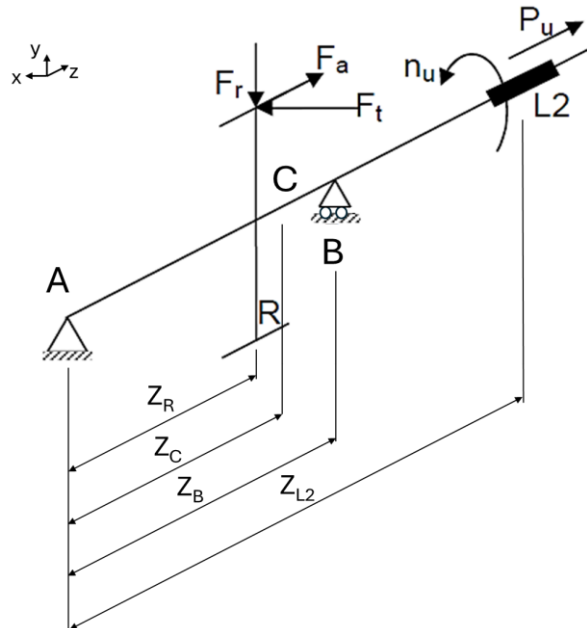


Figure 1. Driven shaft and acting loads

Assume $F_t = 30 \text{ kN}$, $F_a = 8.5 \text{ kN}$, $F_r = 11.1 \text{ kN}$. Dimensions are $Z_r = 100 \text{ mm}$, $Z_b = 270 \text{ mm}$, $Z_{L2} = 350 \text{ mm}$, $R = 155 \text{ mm}$.

The shaft has circular cross section (diameter $d = 75 \text{ mm}$). In C ($Z_c = 120 \text{ mm}$), a shoulder is present, as detailed in Figure 2.

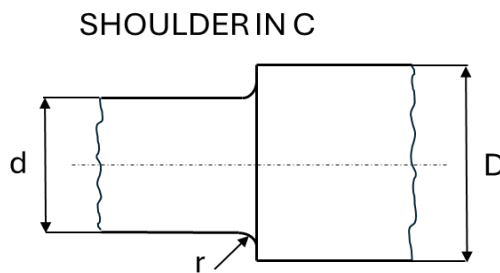


Figure 2. Detail of cross-section C

Consider $D = 85 \text{ mm}$ and $r = 3 \text{ mm}$.

Compute the safety factor against failure.

[SF = 2.5]

E1.1

The secondary shaft (A2) of the parallel shaft gearbox of Figure 1 has two helicoidal gears (GEAR 2 and GEAR 3). In the operational conditions, gears 2 and 3 get in contact with gears 1 and 4, respectively, and the resulting loads are schematically depicted in Figure 2. The secondary shaft is made of steel ($S_y = 315 \text{ MPa}$).

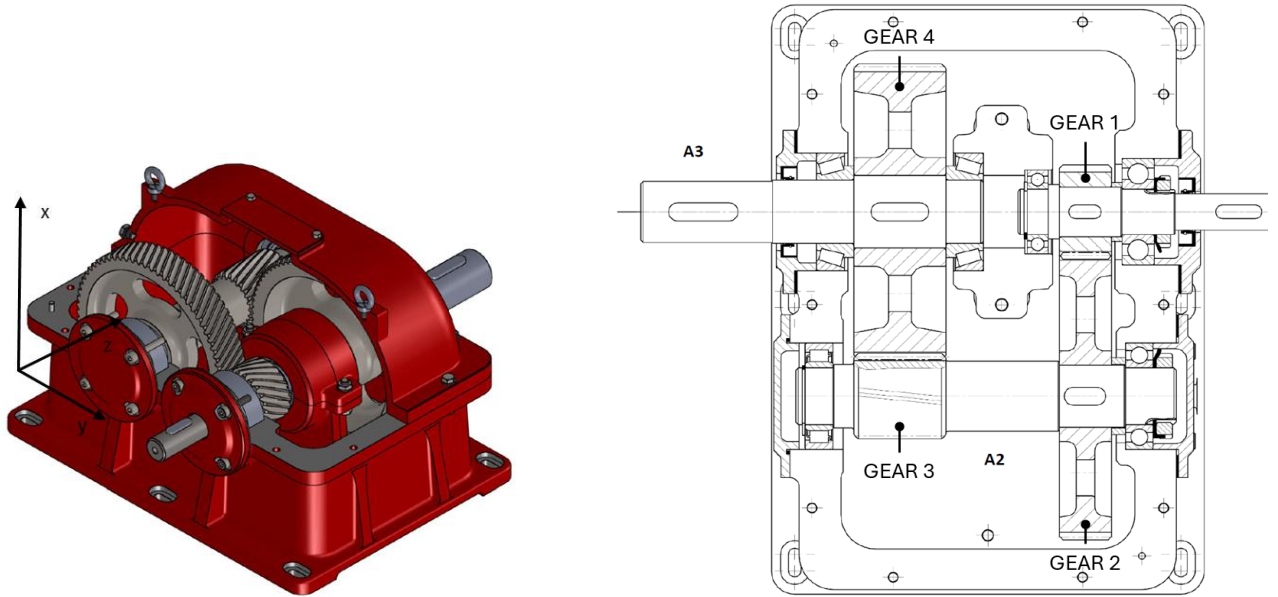


Figure 1. Parallel shaft gearbox.

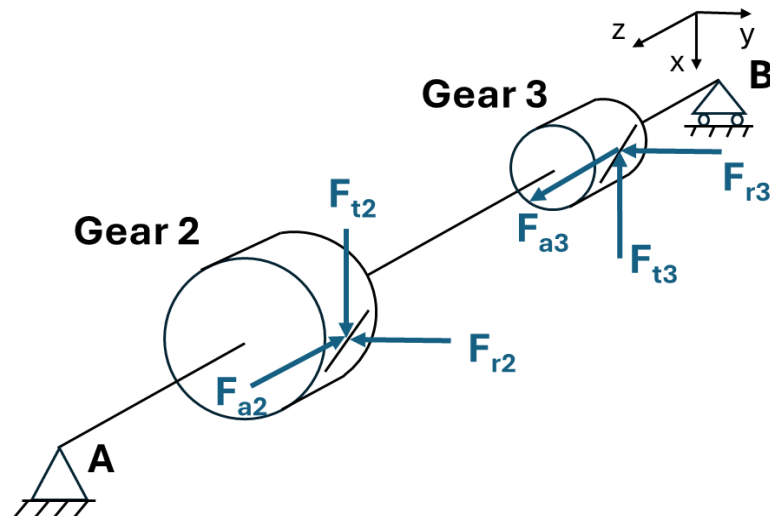


Figure 2. Scheme of loads acting on the secondary shaft.

Assume $F_{a2} = 1000 \text{ N}$, $F_{r2} = 700 \text{ N}$, $F_{t2} = 1600 \text{ N}$, $F_{a3} = 2100 \text{ N}$, $F_{r3} = 2200 \text{ N}$, $F_{t3} = 5760 \text{ N}$.

Details of the shaft diameters are presented in Figure 3. Dimensions are $Z_a = 23 \text{ mm}$, $Z_b = 80 \text{ mm}$, $Z_c = 35 \text{ mm}$. Radius of gear 2 is 62 mm, Radius of gear 3 is 17.5mm.

E1.2

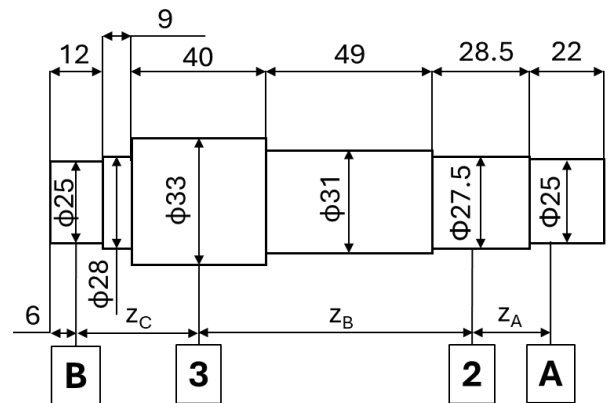
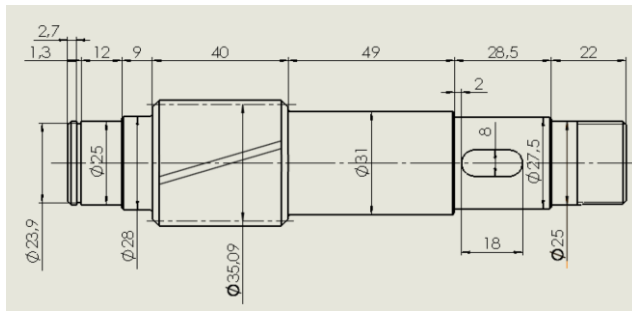


Figure 3. Technical drawing and schematic representation of the secondary shaft

Verify the secondary shaft against yielding.

[SF = 5]

Figure 1 shows the loads acting on the spur gears mounted on the shaft AB, made of S400 steel ($\sigma_Y = 400$ MPa, $\sigma_U = 530$ MPa and $A\% = 22$).

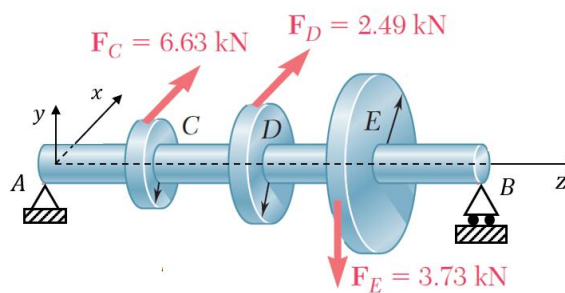
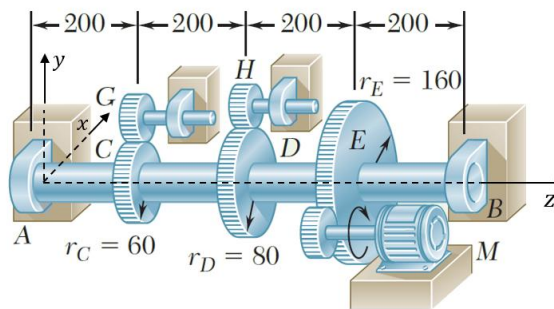


Figure 1.

Compute the minimum shaft diameter that ensures a static factor SF = 2 against initial yielding.

[d = 41 mm]

E1.3

The output shaft S of the parallel-shaft gearbox shown in Figure 1 carries one helical gear G and delivers power to the end user U . The shaft is supported by bearings A and B .

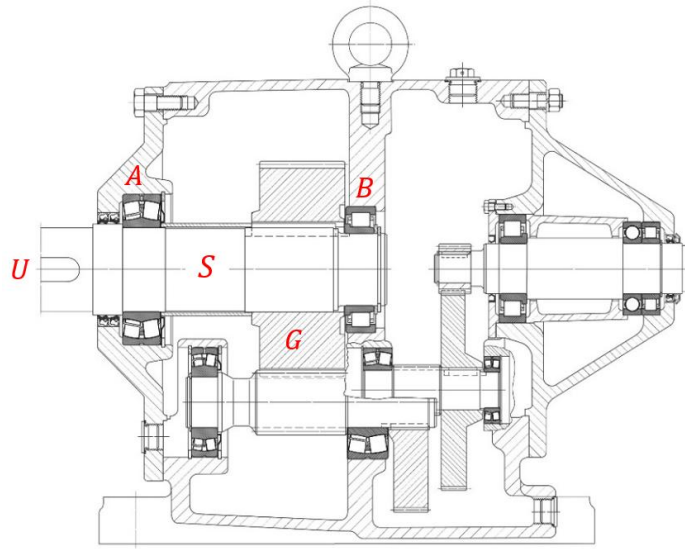


Figure 1.

The main shaft dimensions and the axial positions (along the z -axis) of the user, bearings, and gear connections are given in Figure 2 (where $z_U = 62.5$ mm, $z_A = 185$ mm, $z_G = 366$ mm, $z_B = 435$ mm.)

E1.4

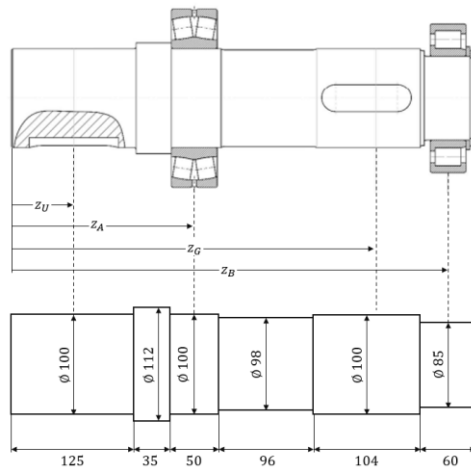


Figure 2.

Assume the loads applied to the helical gear are (see Figure 3): $F_t = 75168$ N, $F_r = 28324$ N, $F_a = 20141$ N, $M_{U_z} = 9020$ Nm (Figure 3) and assume shoulder fillet radii equal to 5 mm.

- Verify shaft S against yielding.
- Verify shaft S against initial yielding.

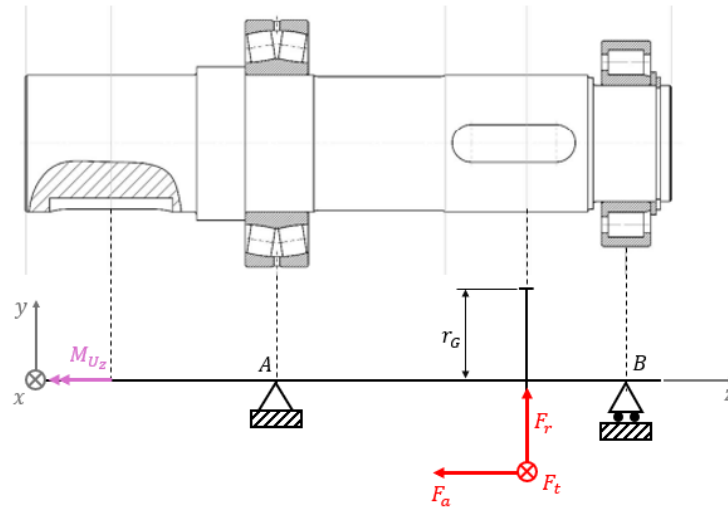


Figure 3.

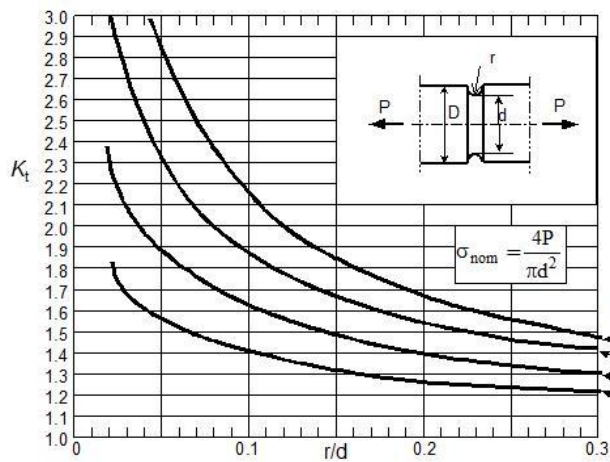
[SF =5; SF=3.7]

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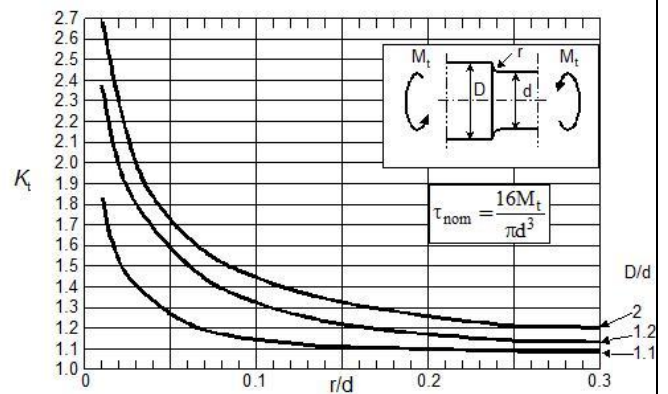
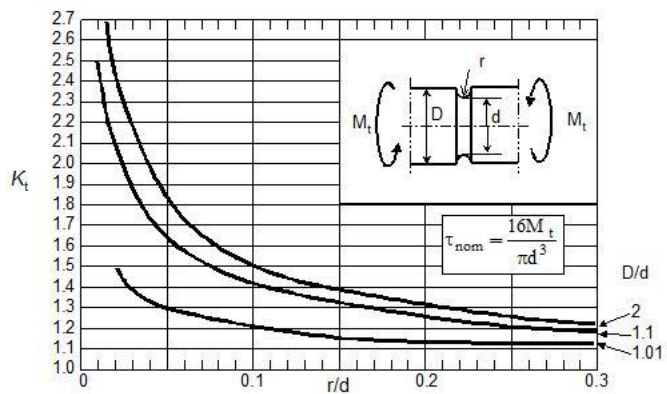
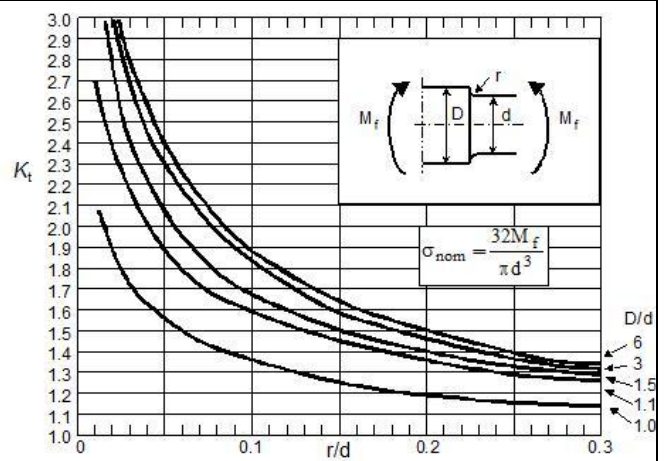
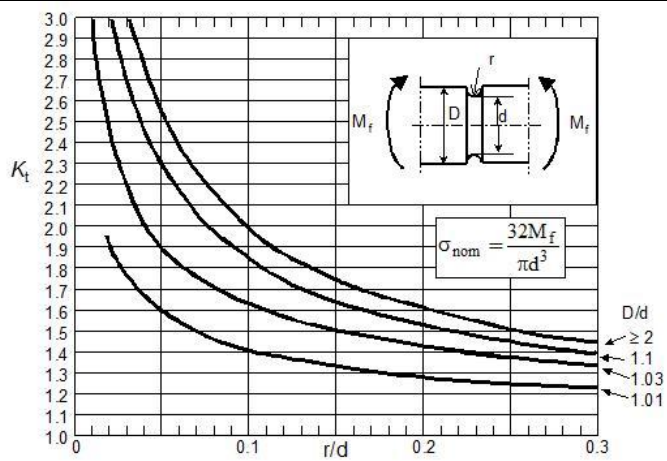
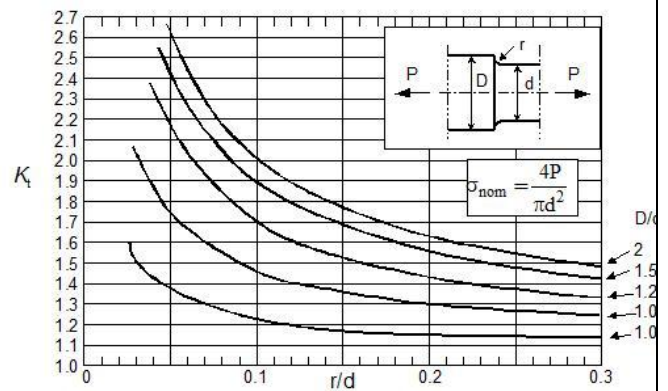
Tutorial #9: 3D structures

Geometrical stress concentration factors K_t

Groove



Shoulder

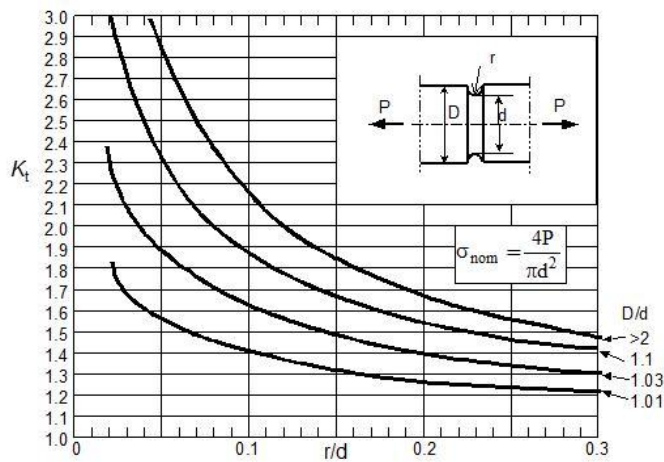


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